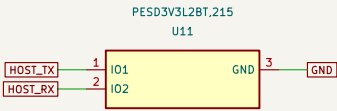
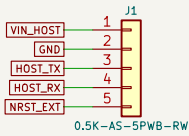
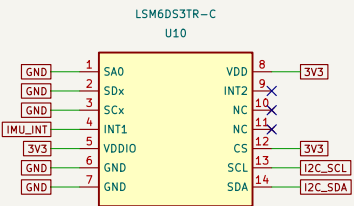
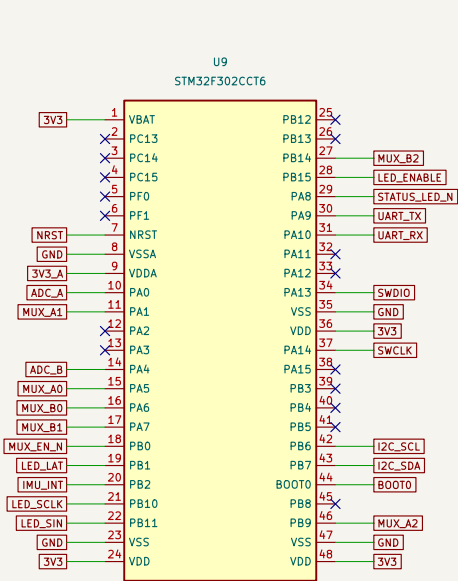


IR GLASSES EVT E6 – CONTROL

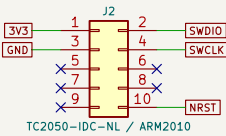


selection

文件: selection.kicad_sch

acquisition

文件: acquisition.kicad_sch



power

文件: power.kicad_sch

left

文件: left.kicad_sch

right

文件: right.kicad_sch

decoupling

文件: decoupling.kicad_sch

Protection and status

文件: protection.kicad_sch

Prototype – mechanical and optical validation required

cc_glasses

Sheet: /
File: ir_glasses_EVT_E6_F302.kicad_sch

Title: IR Glasses – control

Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-18

Rev: EVT E6 F302 R2

Id: 1/9

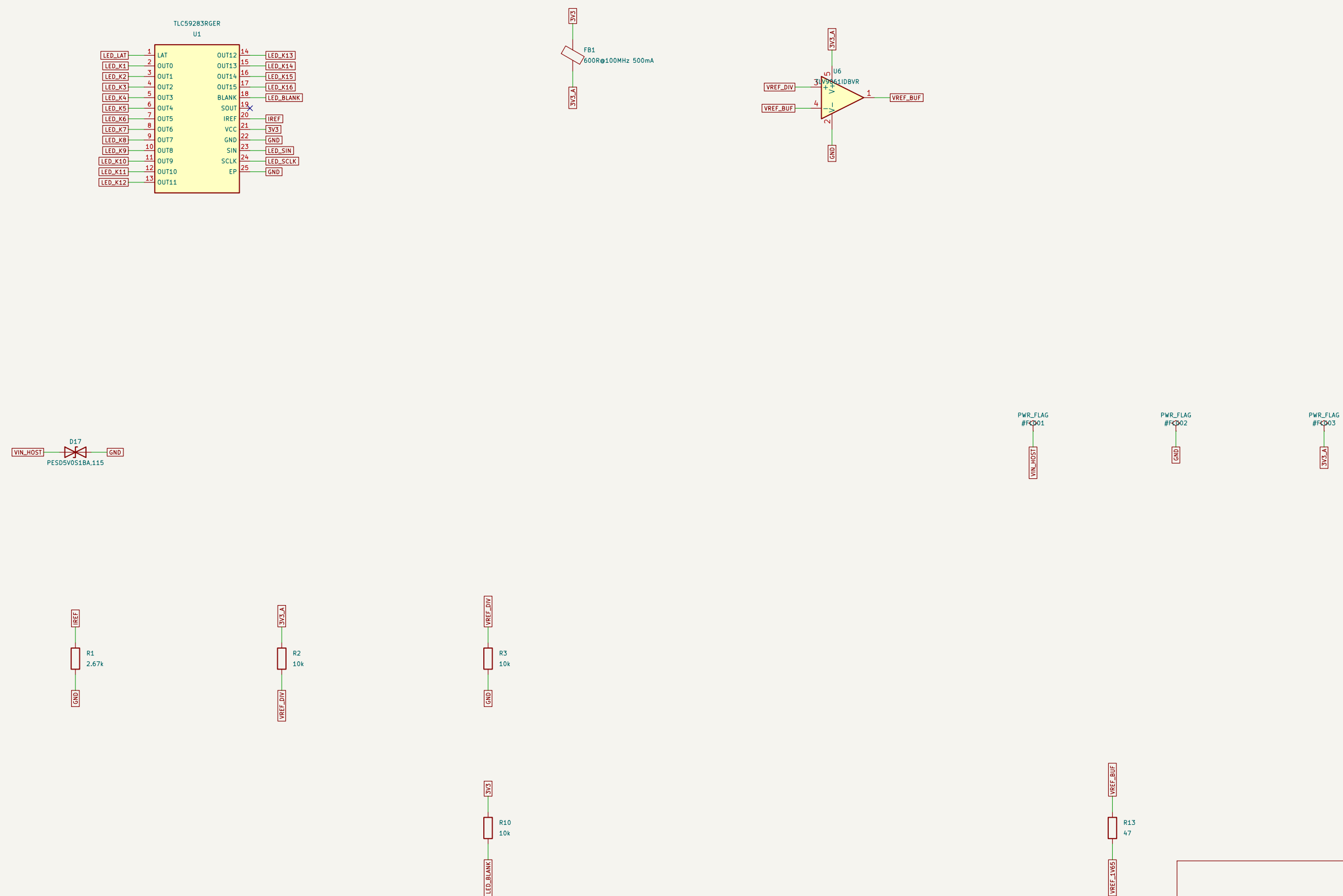
STM32F302CCT6: dual ADC, 30/24 fps, BOOT0 has R5 pull-down; SWCLK is independent.

input_protection

文件: input_protection.kicad_sch

20 mA/ch; one LED at a time. BLANK pulled high during reset.

TIA: PD cathode at virtual ground, anode at GND; output rises with light.



Prototype – mechanical and optical validation required

cc_glasses

Sheet: /power/
File: power.kicad_sch

Title: IR Glasses – power

Size: A3

Date: 2026-09-18

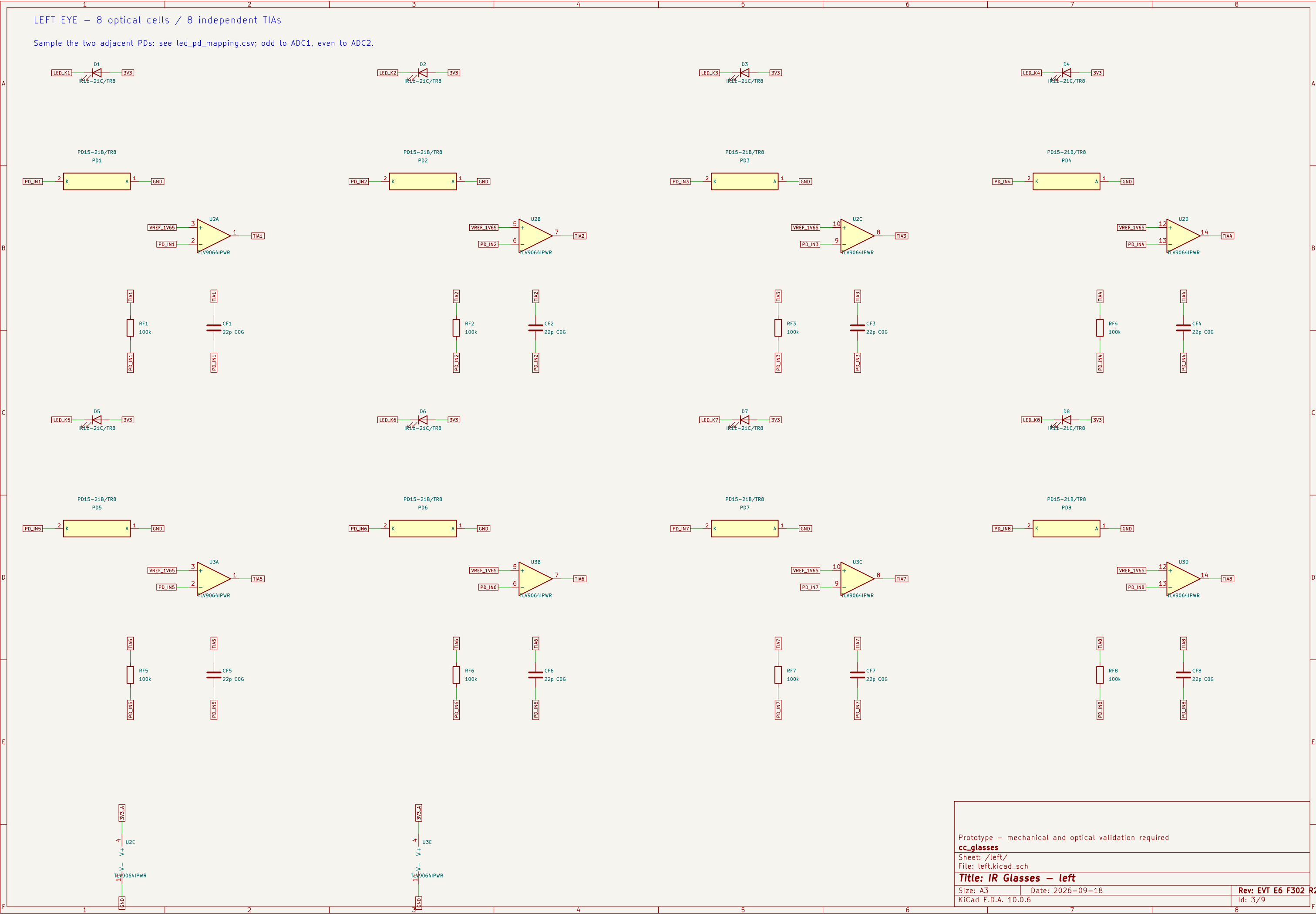
Rev: EVT E6 F302 R2

Size: A5	
KiCad E.D.A. 10.0.6	

Id: 2/9

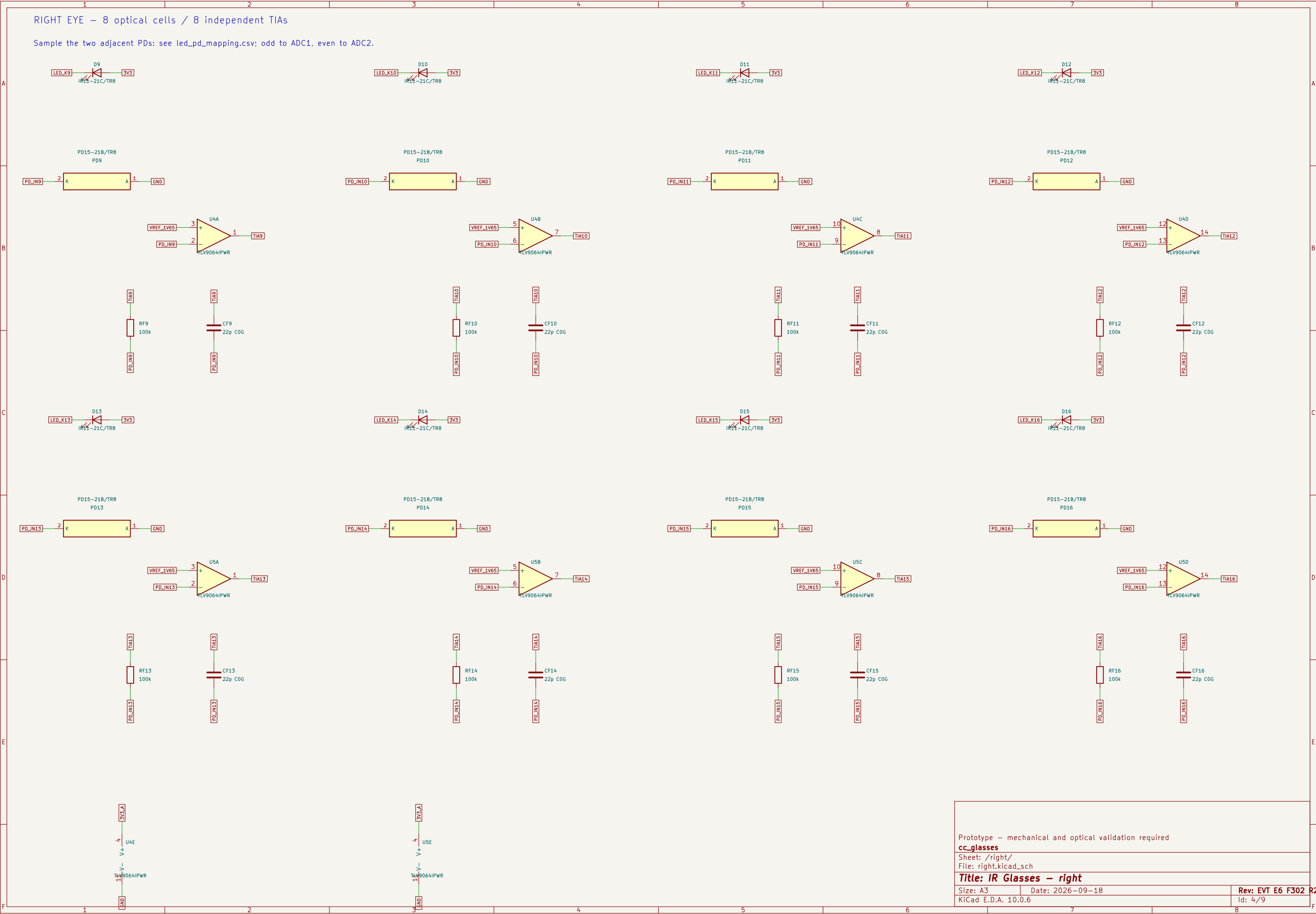
LEFT EYE – 8 optical cells / 8 independent TIAs

Sample the two adjacent PDs: see led_pd_mapping.csv; odd to ADC1, even to ADC2.

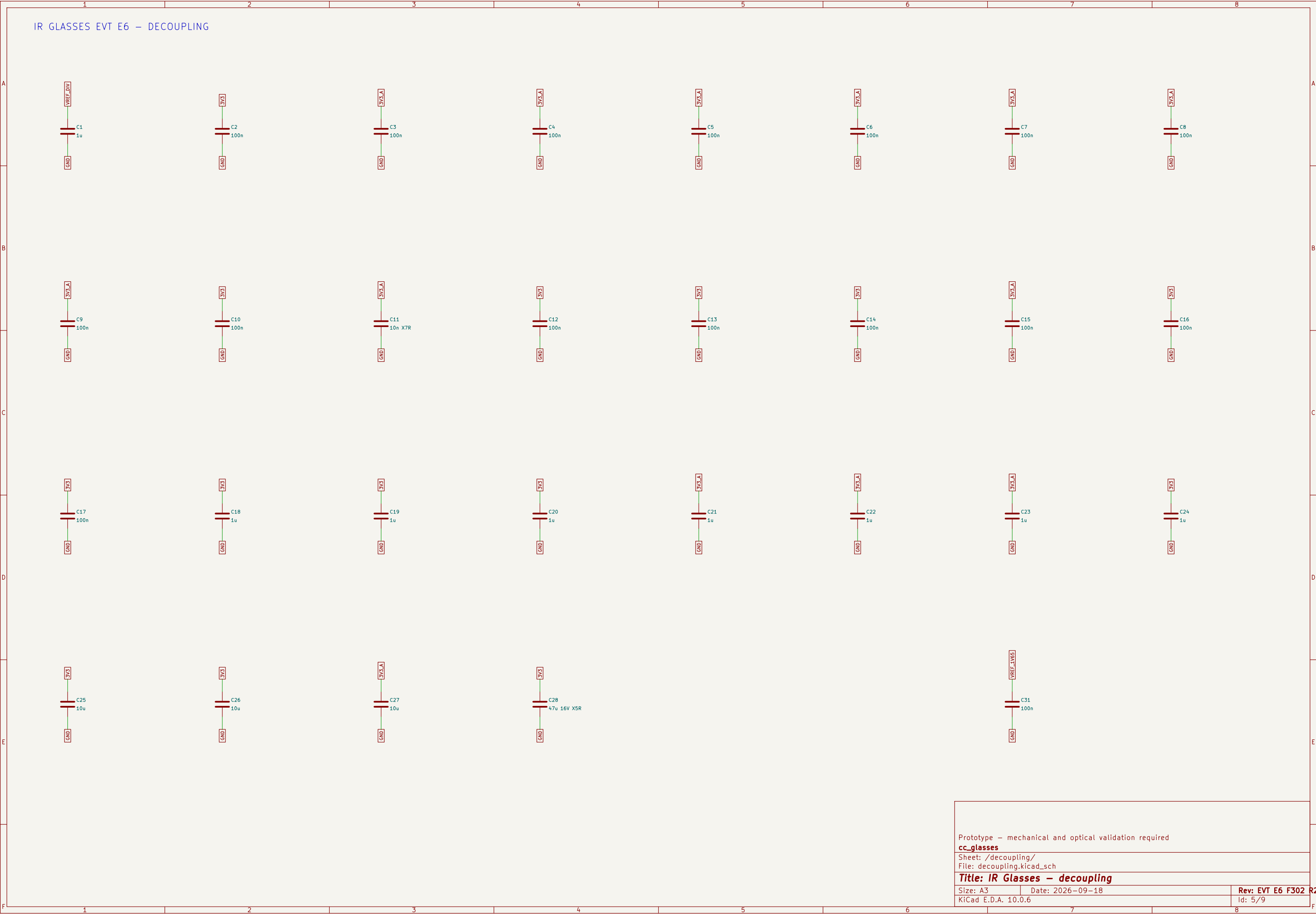


RIGHT EYE – 8 optical cells / 8 independent TIAs

Sample the two adjacent PDs: see led_pd_mapping.csv; odd to ADC1, even to ADC2.

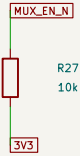
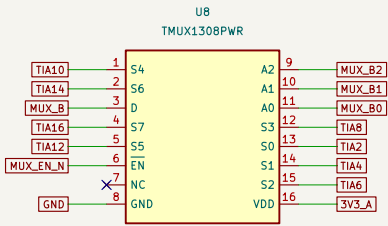
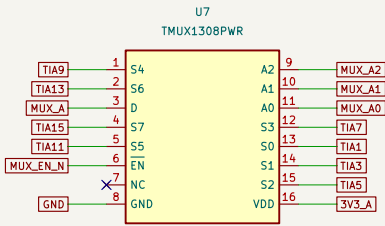


Prototype – mechanical and optical validation required		
cc_glasses		
Sheet: /right/		
File: right.kicad_sch		
Title: IR Glasses – right		
Size: A3	Date: 2026-09-18	Rev: EVT E6 F302 R2
KiCad E.D.A. 10.0.6		Id: 4/9





IR GLASSES EVT E6 – ADJACENT ODD / EVEN PD SELECTION



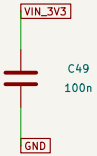
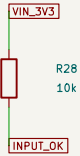
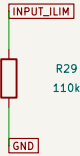
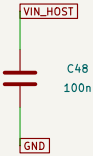
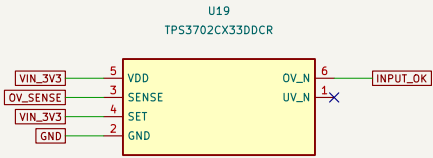
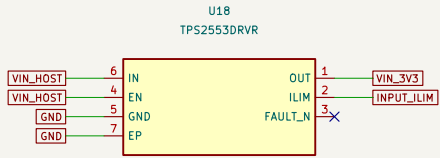
U7 selects odd PDs; U8 selects even PDs. LED_ID selects its two adjacent PDs using the fixed table.

EN_N is active low; R27 pulls high to disable both paths during reset.

Switch only with LED_ENABLE low. MUX address = (PD_ID – 1) / 2, integer division.



INPUT PROTECTION – 3.3 V NOMINAL / CORRECT POLARITY



J1 -> U18 (110k current limit) -> VIN_3V3 -> U14 -> 3V3.

U19 TPS3702CX33: SET=VIN; R18=200R / R19=10k; OV_N -> U14 EN.

U14 OVLO=GND. EN low disables; recovery follows the C38=22nF startup path.

Nominal OV trip 3.501V / release 3.481V. Input: 3.3V +/-3%, correct polarity.

No guaranteed fast surge clamp: comparator and EN response have delays.

Bench validation required: startup, hot-plug, recovery, short circuit and temperature.